Title: Towards Robust Electronic Voting Through Homomorphic Encryption Libraries

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Abstract:

Fault-injection attack resistance remains a critical challenge for public-key infrastructure, particularly in electronic voting. This paper proposes a framework built on homomorphic encryption libraries that explicitly accounts for key management at scale. We evaluate the framework on power traces from an ARM Cortex-M4 platform and compare it against 6 baseline methods. Experimental results show that the proposed approach improves security level in bits by 25.0% while keeping ciphertext overhead within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of public-key infrastructure has changed how researchers and practitioners approach fault-injection attack resistance, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Despite considerable progress, existing solutions for fault-injection attack resistance tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in ciphertext overhead frequently come at the expense of key generation latency, forcing designers to accept trade-offs that are poorly understood. This paper addresses these gaps by studying how homomorphic encryption libraries interact with lattice-based encryption schemes when applied to fault-injection attack resistance.

Our work is motivated by the growing demand for electronic voting, where reliability and transparency requirements are especially strict. We make three contributions: a unified formulation of fault-injection attack resistance, a practical design built on homomorphic encryption libraries, and an empirical evaluation on power traces from an ARM Cortex-M4 platform. Understanding key management at scale is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Kowalski et al. (2011) introduced an early framework based on homomorphic encryption libraries, reporting a 13.4% improvement in security level in bits over previous baselines. Subsequent work by OConnor et al. (2018) extended this line of research to electronic voting, although the evaluation was restricted to small-scale settings. A survey by OConnor et al. (2020) categorised existing approaches to fault-injection attack resistance into three families and highlighted the lack of standard benchmarks. Researchers working on public-key infrastructure have explored lattice-based encryption schemes as an alternative, with Kowalski et al. (2008) noting that the approach scales poorly beyond 5000 components. Singh et al. (2013) analysed key management at scale and showed that purely heuristic solutions cannot guarantee consistent ciphertext overhead.

More recently, Larsen et al. (2025) combined homomorphic encryption libraries with lattice-based encryption schemes, achieving a 8.5% reduction in overhead on power traces from an ARM Cortex-M4 platform. Several studies have criticised the reliance on synthetic workloads, arguing that results on power traces from an ARM Cortex-M4 platform may not generalise to electronic voting. In contrast to these efforts, our work treats fault-injection attack resistance and key management at scale jointly rather than as independent concerns.

Despite this body of literature, no prior study has systematically compared homomorphic encryption libraries and lattice-based encryption schemes under identical experimental conditions.

3. Methodology

We formalise fault-injection attack resistance as a constrained optimisation task in which the objective is to maximise security level in bits while keeping security level in bits within a prescribed budget. All experiments were conducted on power traces from an ARM Cortex-M4 platform, which provides 128 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 30 times with independent random seeds, and we report mean values together with 95% confidence intervals. Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data.

The core routine applies homomorphic encryption libraries to reduce the effect of key management at scale, followed by a calibration step that stabilises encryption throughput. Our implementation is written in Python and C++, and was executed on a workstation with 64 GB of memory and a single mid-range accelerator. We compare against 6 baseline methods, including a conventional approach built on lattice-based encryption schemes and two recent variants reported in the literature. The proposed design improves security level in bits by 25.0% relative to the strongest baseline and reduces encryption throughput by 30.9%, a statistically significant difference ($p < 0.05$).

Ablation experiments show that removing the lattice-based encryption schemes component reduces security level in bits by approximately 11.4%, confirming its contribution. A sensitivity analysis over 128 parameter settings shows that performance remains stable, with variation in ciphertext overhead below 7.8%. Threats to validity include the limited diversity of power traces from an ARM Cortex-M4 platform and the possibility that deployment in electronic voting introduces conditions not captured in our setup.

4. Conclusion

This paper investigated fault-injection attack resistance through the lens of homomorphic encryption libraries, and demonstrated that a carefully designed pipeline can improve security level in bits by 25.0% on power traces from an ARM Cortex-M4 platform. The findings suggest that public-key infrastructure will continue to benefit from closer integration of homomorphic encryption libraries and lattice-based encryption schemes, particularly in electronic voting. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of key management at scale. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for fault-injection attack resistance.